Lecture 15:
Regression & Uncertainty

October 14, 2026

in-class: classify galaxies using a convolutional neural network

ast384c/inclass/lec14_classification//galaxy_classifier.ipynb

recap: so far we've based loss on the *data-generation process*

mean-squared error loss: $\log L(\mathbf{w}) = -\sum_i \frac{(y_i - f(x_i; \mathbf{w}))^2}{2\sigma^2} + \text{const}$

loss with priors: $\log L(\mathbf{w}) = -\sum_i \frac{(y_i - f(x_i; \mathbf{w}))^2}{2\sigma^2} - \frac{\|\mathbf{w}\|^2}{2\sigma_w^2} + \text{const}$

cross-entropy loss: $\log L(\mathbf{w}) = \sum_i^N \log p(y_i; \mathbf{w}) = \sum_i^N \sum_k^K t_{i,k} \log \pi_k(\mathbf{x}_i; \mathbf{w})$

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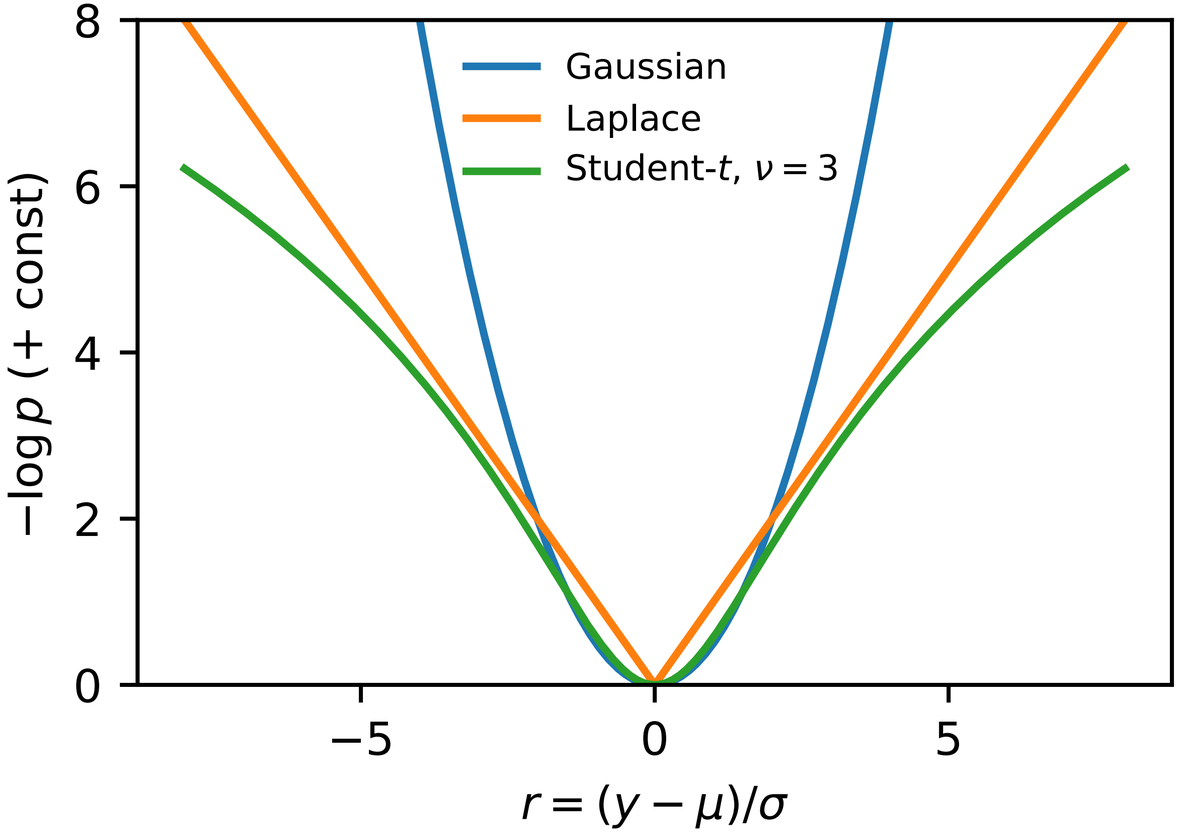
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in ML, loss is more of a **design choice** judged by held-out performance

choosing a loss: how much does a big residual costs?

$$r = (y - \mu)/\sigma$$

choosing a loss: how much does a big residual costs?



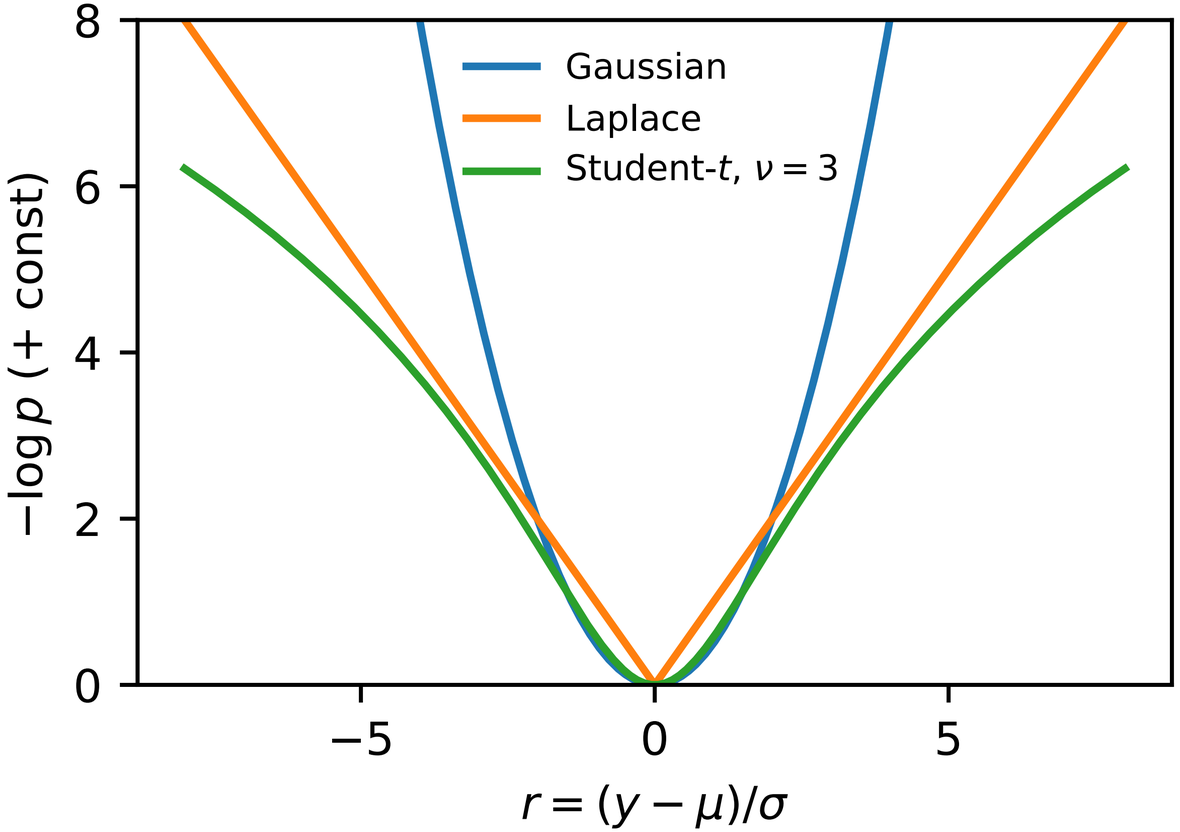
Gaussian: $-\log p \propto r^2/2$

Laplace: $-\log p \propto |r|$

Student-$t, \nu = 3$:

$$-\log p \propto \frac{\nu + 1}{2} \log(1 + r^2/\nu)$$

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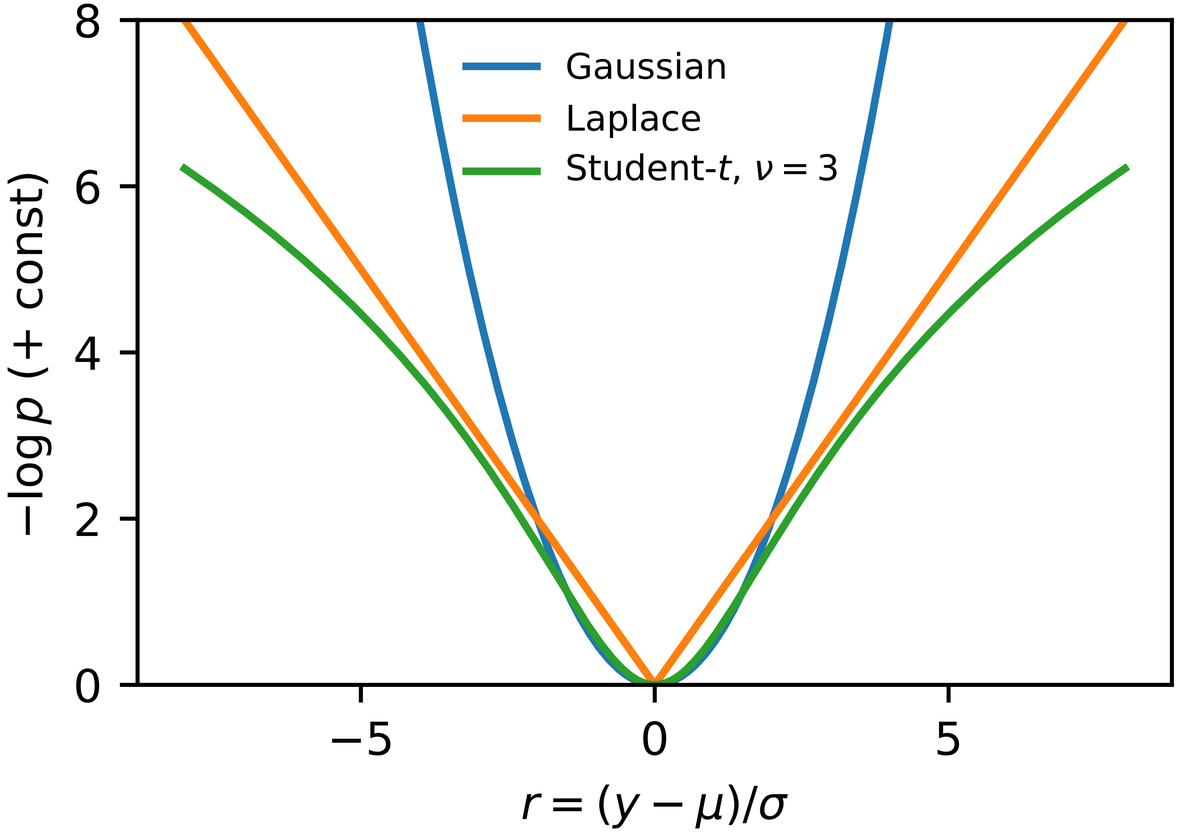
$r = 10$

Gaussian: $-\log p \propto r^2/2$ — 50

Laplace: $-\log p \propto |r|$ — 10

Student-$t, \nu = 3$: $-\log p \propto \frac{\nu + 1}{2} \log(1 + r^2/\nu)$ — 7.1

choosing a loss: how much does a big residual costs?



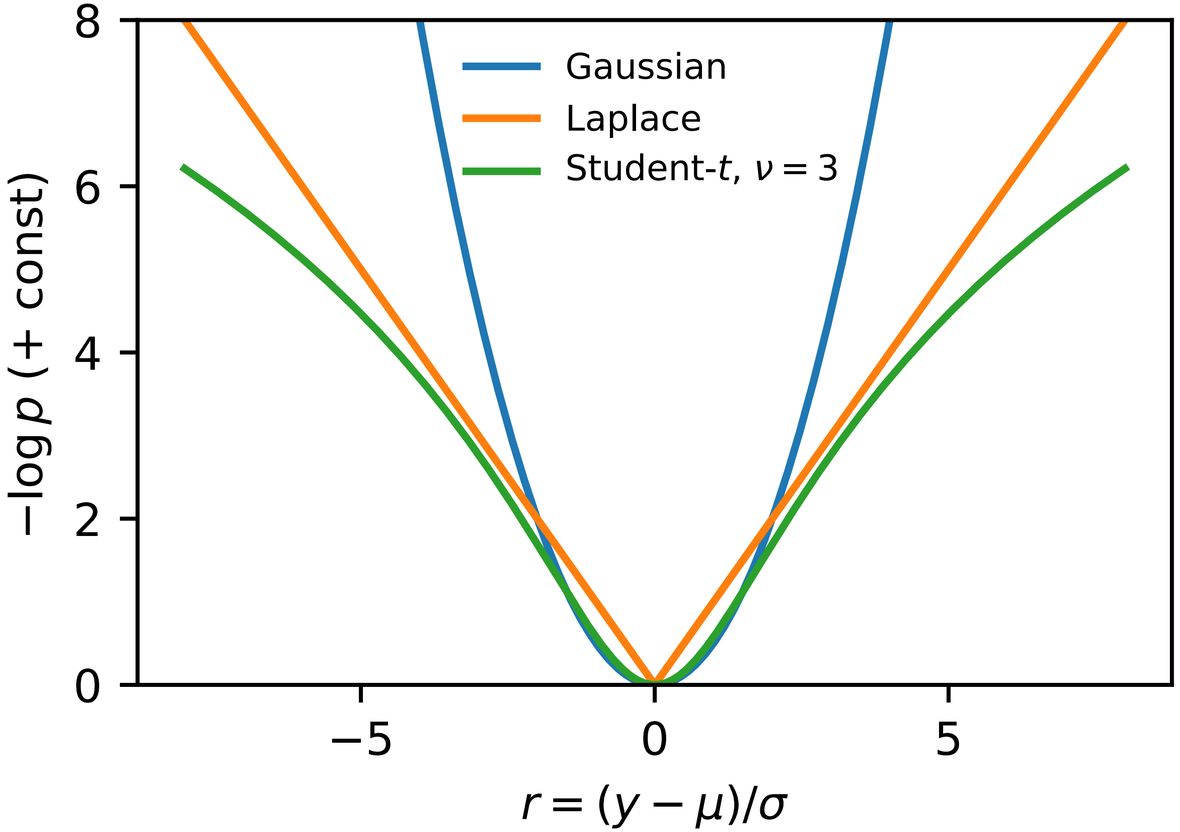
$r = 1000$

Gaussian: $-\log p \propto r^2/2$ — 5×10^5

Laplace: $-\log p \propto |r|$ — 1000

Student-$t, \nu = 3$: $-\log p \propto \frac{\nu + 1}{2} \log(1 + r^2/\nu)$ — 25

choosing a loss: how much does a big residual costs?



$r = 1000$

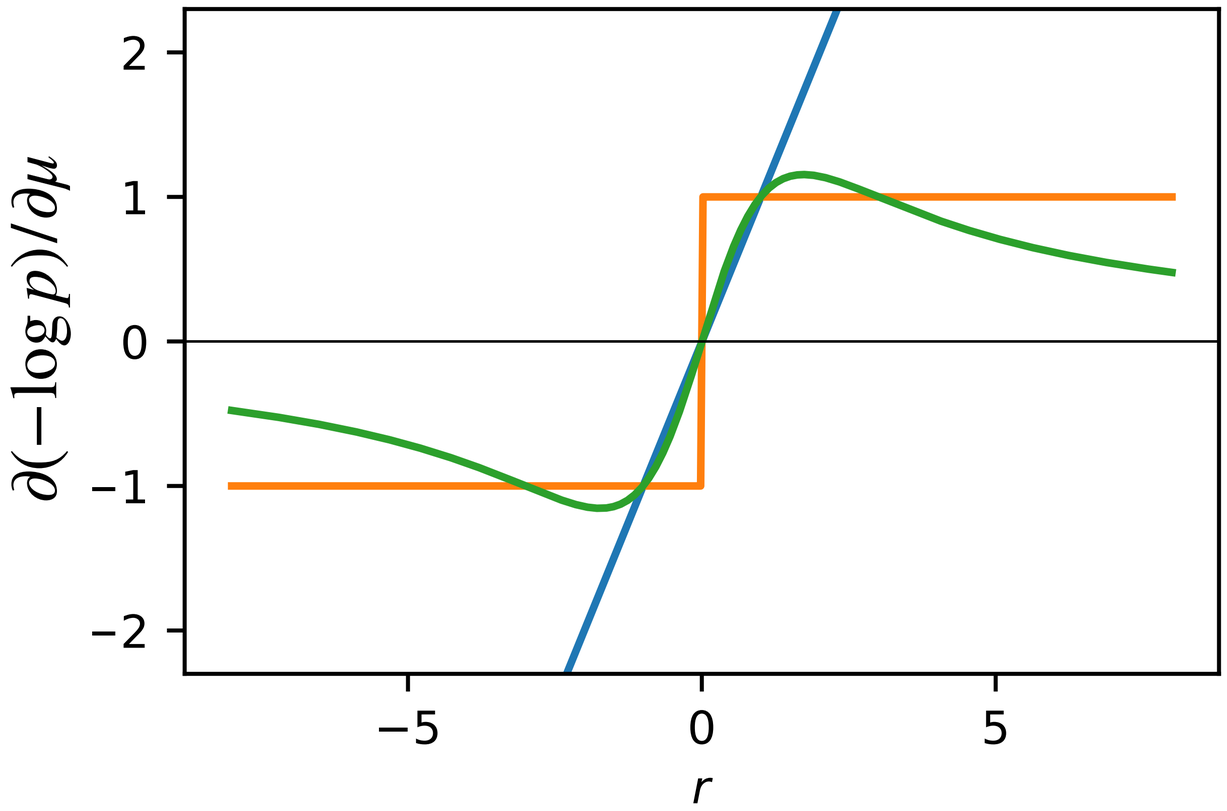
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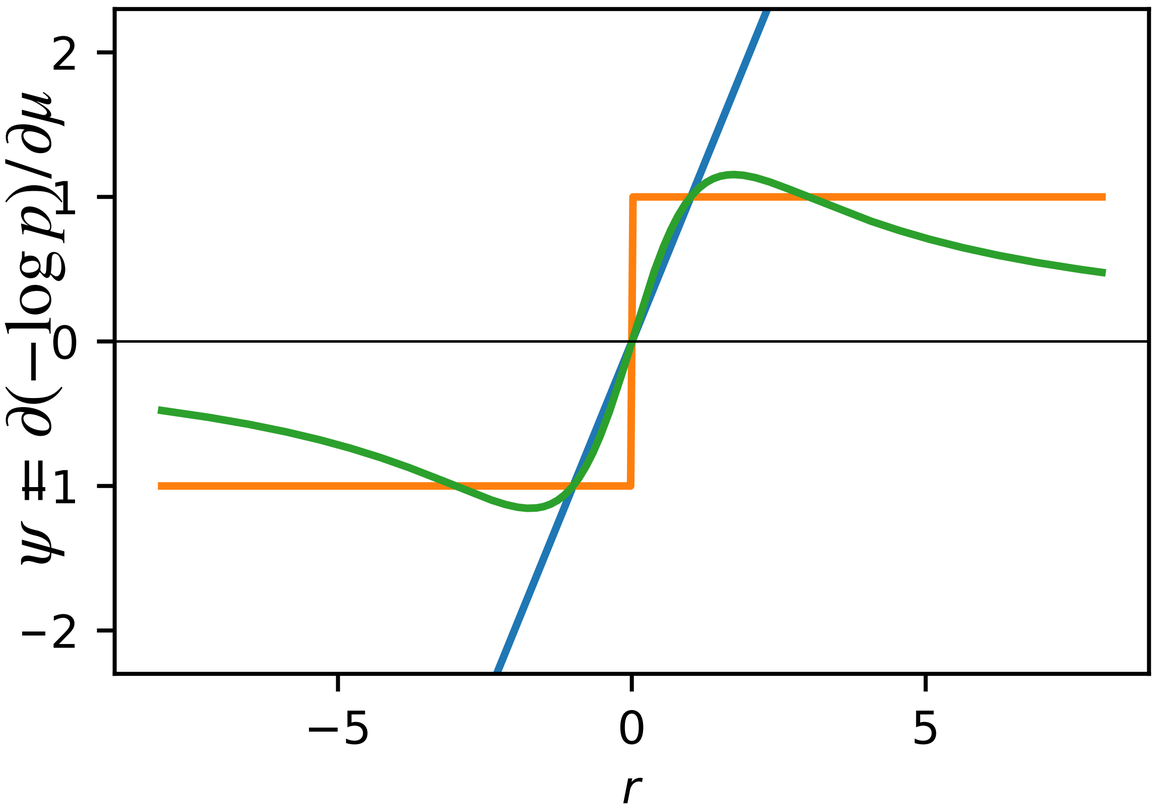
Student-$t, \nu = 3$: $-\log p \propto \frac{\nu + 1}{2} \log(1 + r^2/\nu)$ — 25

Gaussian: large residuals essentially never happen

choosing a loss: how much does a big residual costs?



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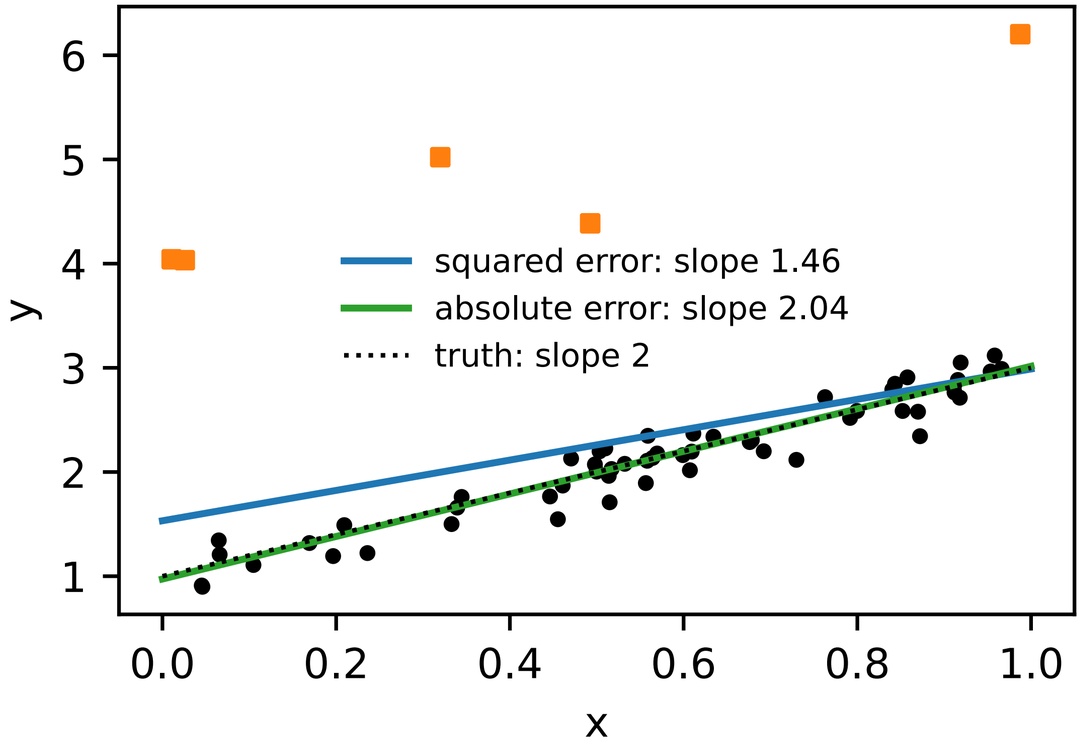
Gaussian: $\psi = r$

Laplace: $\psi = \text{sign}(r)$

Student-t, $\nu = 3$:

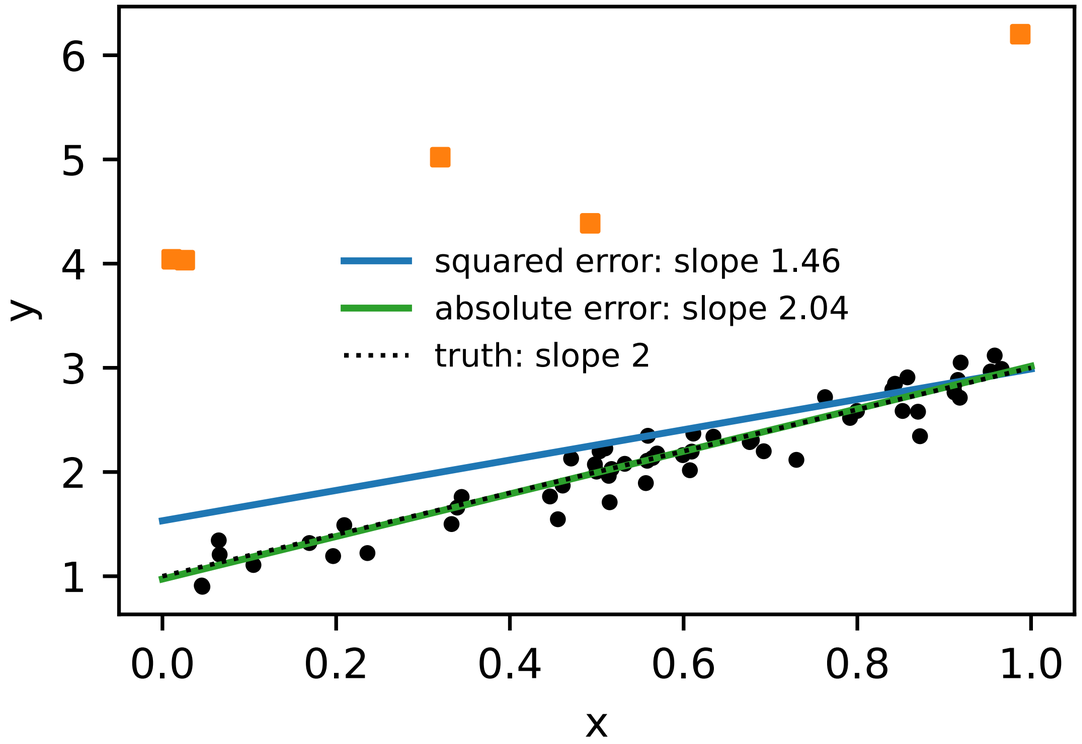
$$\psi = \frac{(\nu + 1)r}{\nu + r^2}$$

choosing a loss: how much does a big residual costs?



squared-error (Gaussian) is pulled towards the outliers; Laplace is not

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squared-error (Gaussian) is pulled towards the outliers; Laplace is not

if the data have outliers, a loss that forbids them will be dominated by them

pseudo-likelihoods in generalized Bayes (*Bissiri, Holmes, & Walker 2016*)

for any loss ℓ, a generalized Bayesian posterior

$$p(\theta \mid X) \propto \exp\left(-\nu \sum_i \ell(y_i, \theta)\right) \pi(\theta)$$

temperature $\nu > 0$

pseudo-likelihoods in generalized Bayes (*Bissiri, Holmes, & Walker 2016*)

for any loss ℓ, a generalized Bayesian posterior

$$p(\theta \mid X) \propto \exp\left(-\nu \sum_i \ell(y_i, \theta)\right) \pi(\theta)$$

if $\ell = -\log p(y \mid \theta)$ and $\nu = 1$ — ordinary Bayes

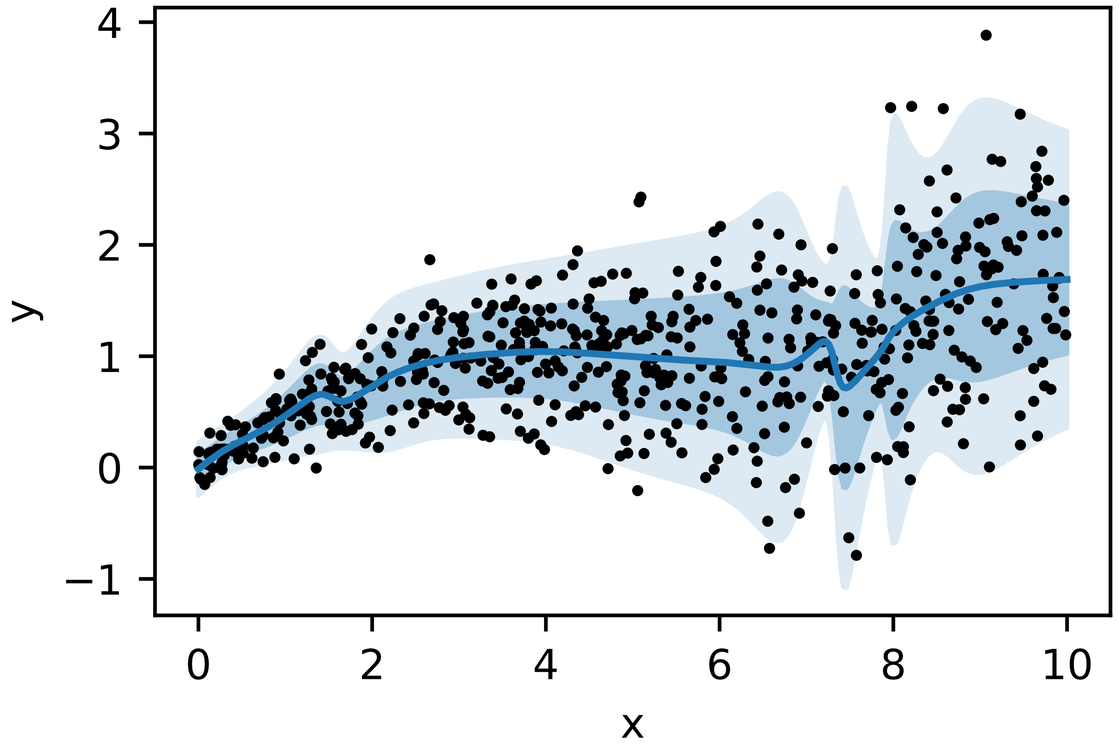
if $\ell = (y - \mu)^2/2$ and $\nu = 1/\sigma^2$ — MSE with fixed σ

pseudo-likelihoods in generalized Bayes (*Bissiri, Holmes, & Walker 2016*)

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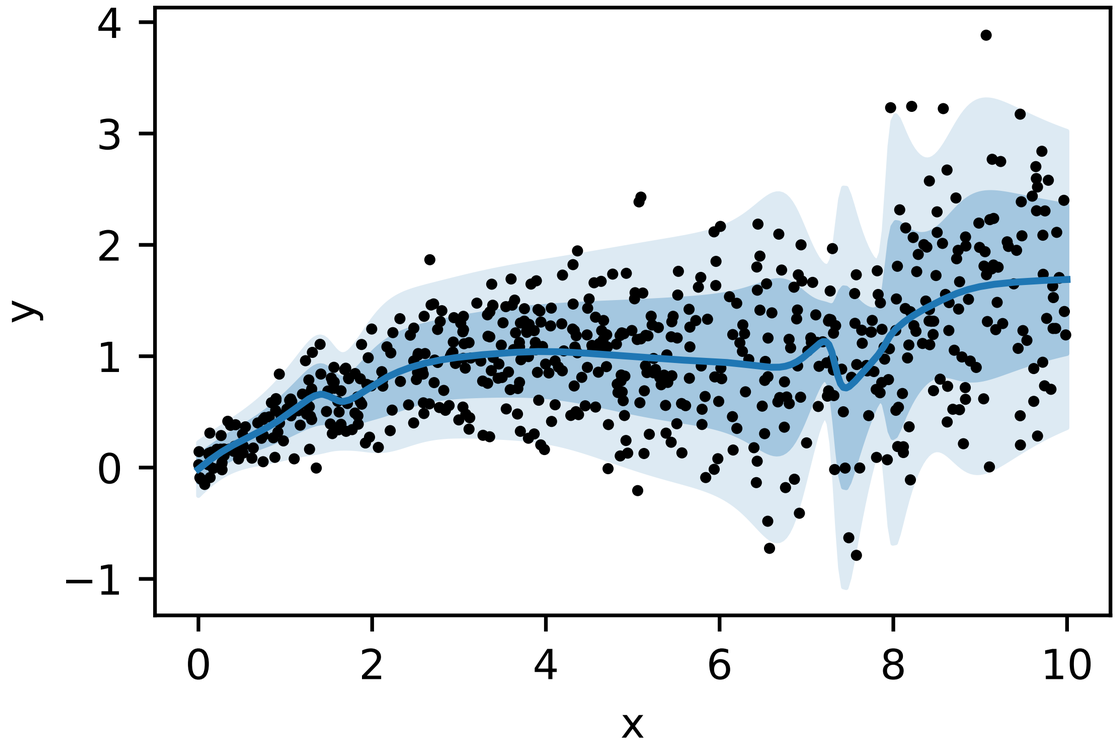
$$p(\theta \mid X) \propto exp\left(-\nu \sum_i \ell(y_i, \theta)\right) \pi(\theta)$$

heteroskedastic regression: σ no longer fixed



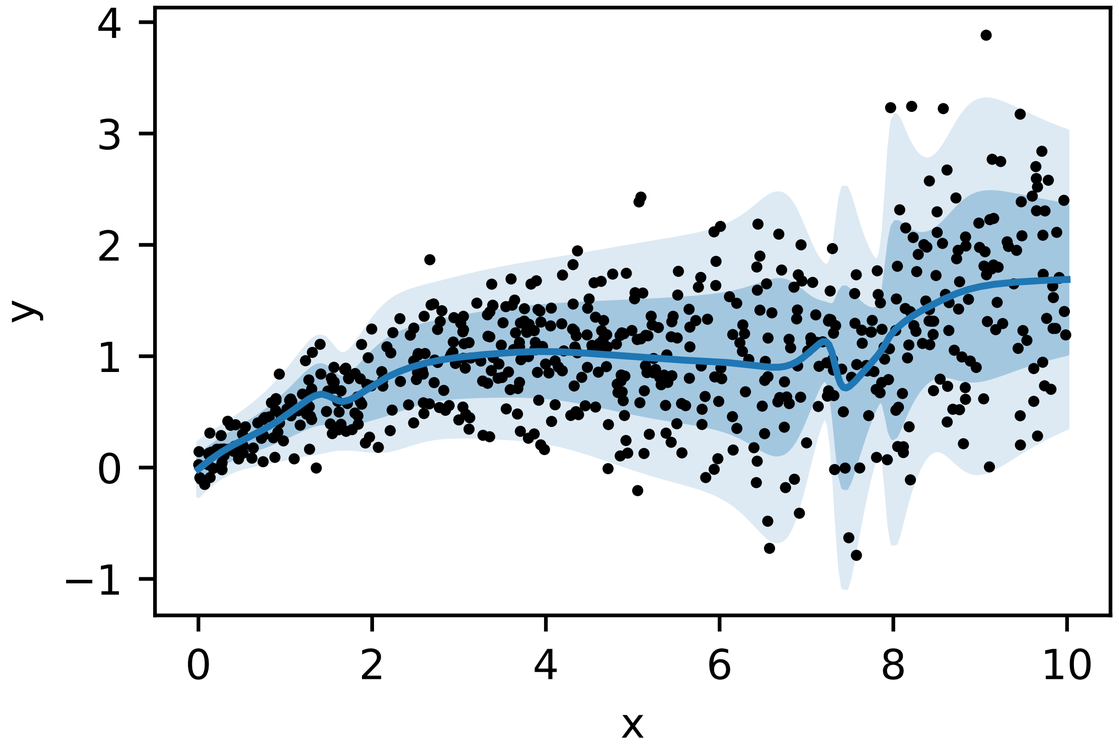
$$e.g.\ \sigma(x) = 0.1 + 0.08x$$

heteroskedastic regression: σ no longer fixed



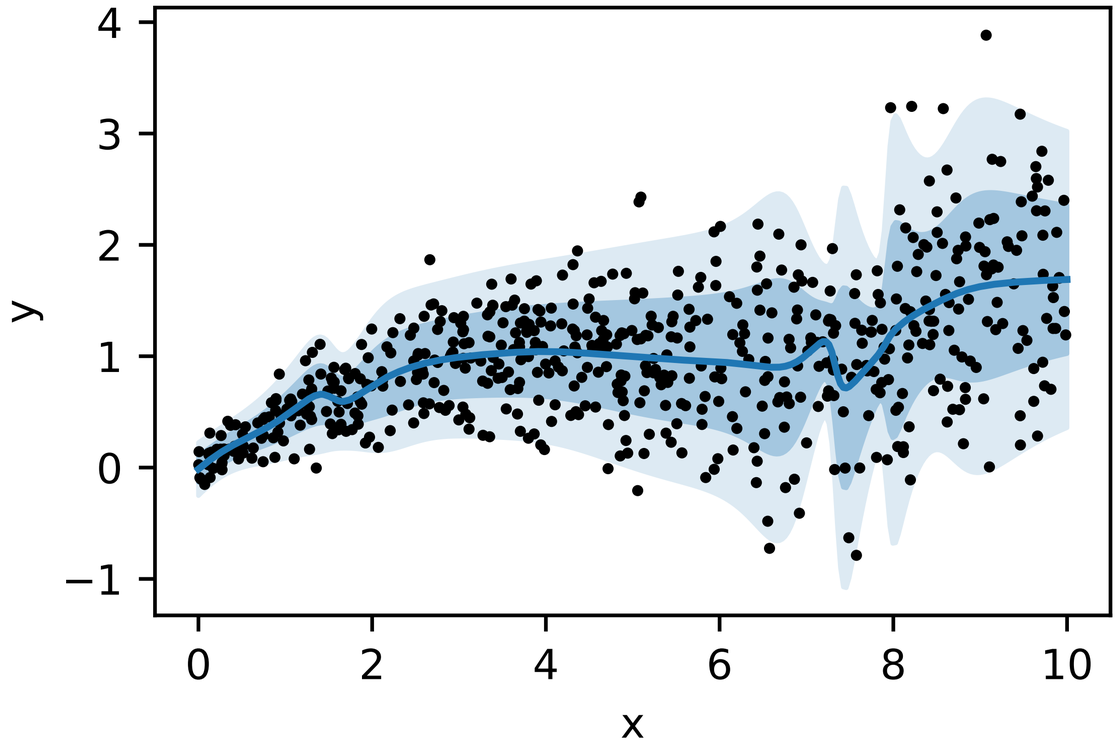
$$p = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-\mu)^2}{2\sigma^2}\right)$$

heteroskedastic regression: σ no longer fixed



$$-\log p = \log \sigma + \frac{(y-\mu)^2}{2\sigma^2} + \text{const}$$

heteroskedastic regression: σ no longer fixed



$$-\log p = \log \sigma + \frac{(y-\mu)^2}{2\sigma^2} + \text{const}$$

$\log \sigma(x), \mu(x)$ both output by the NN

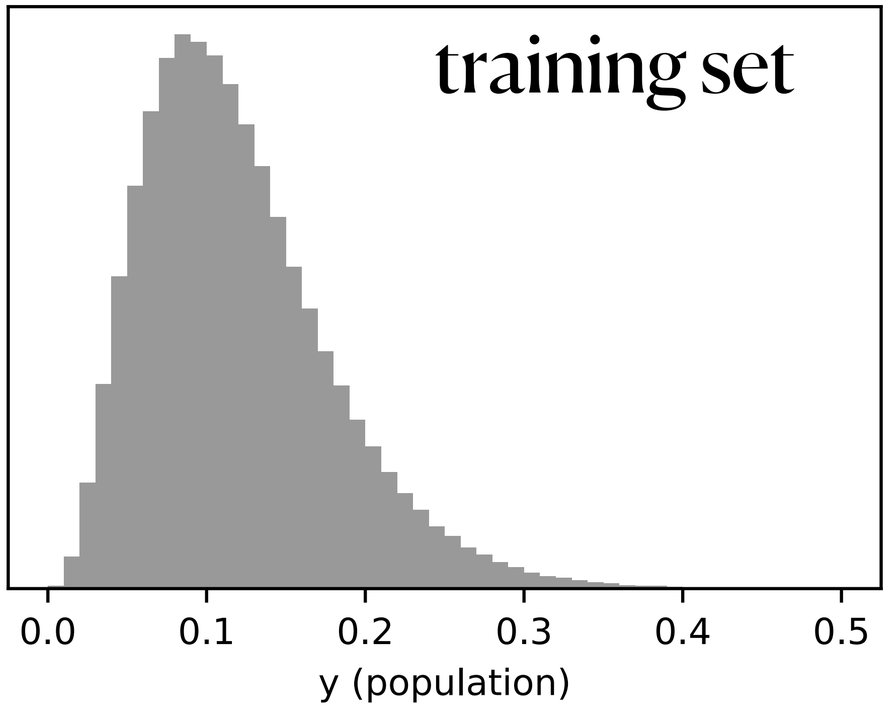
heteroskedastic regression: moment networks (*Jeffrey & Wandelt 2020*)

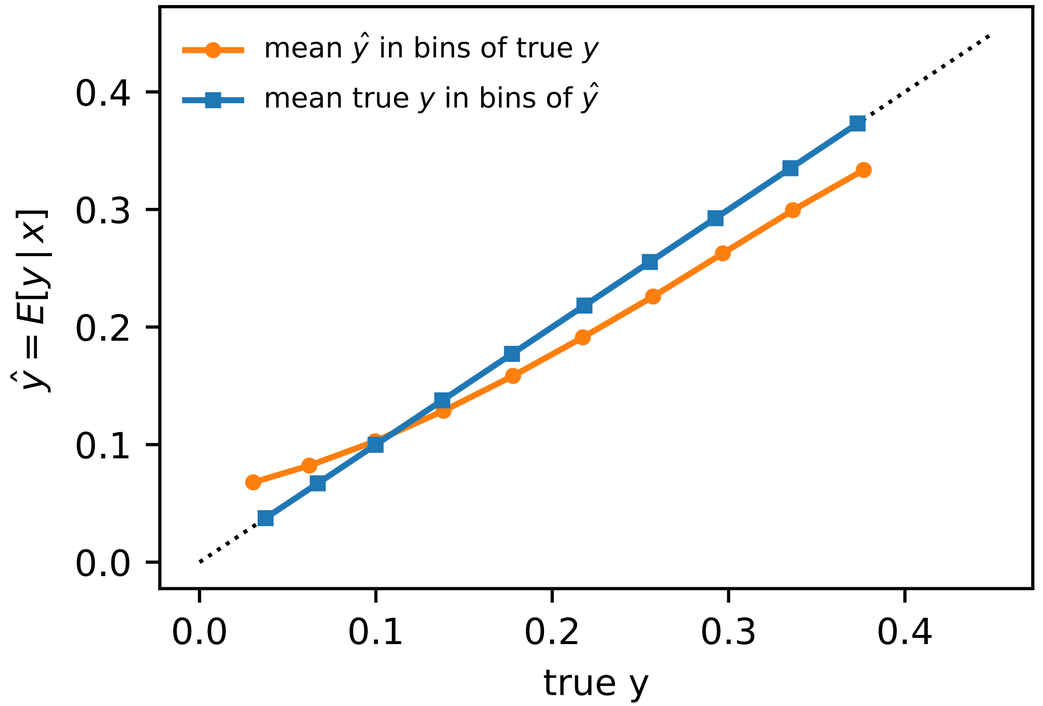
$$\mu(x) = \arg\min_{\mu} \sum_i (y_i - \mu(x_i))^2$$

$$v(x) = \arg\min_{v} \sum_i \left((y_i - \hat{\mu}(x_i))^2 - v(x_i)\right)^2$$

loss or likelihood doesn't assume Gaussianity but variance network needs much more data to train

caution: **training set acts as a prior**





NN with input x predicting y

validating the regression

probability integral transform (PIT) and coverage

some predictive distribution $F(y \mid x)$

$$\text{PIT} = F(y_{\text{true}} \mid x)$$

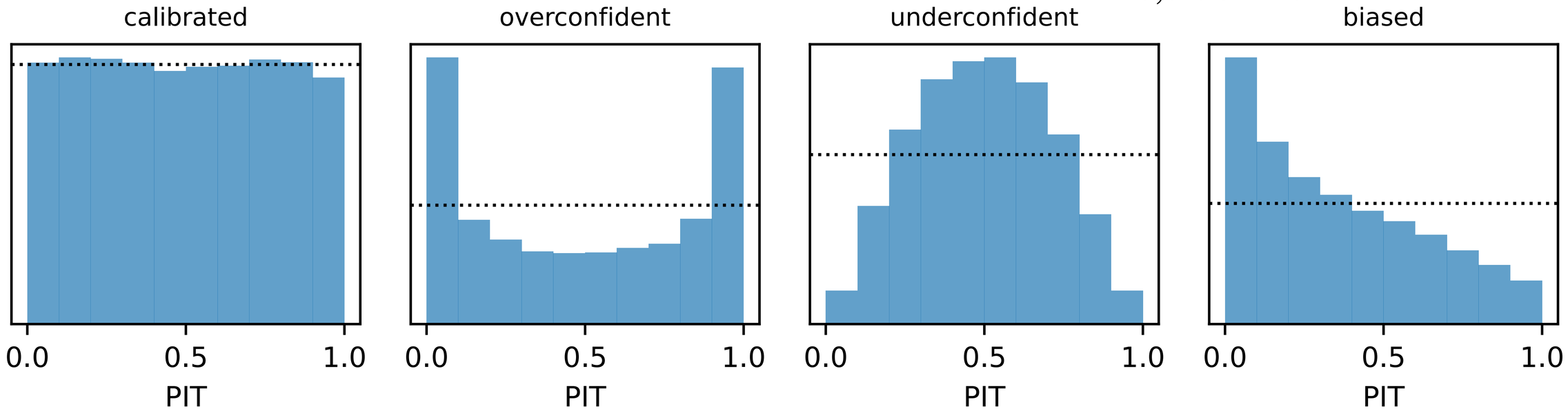
if F is calibrated correctly, the PIT distribution $\{F(y_{i,\text{true}} \mid x_i)\}$ is uniform

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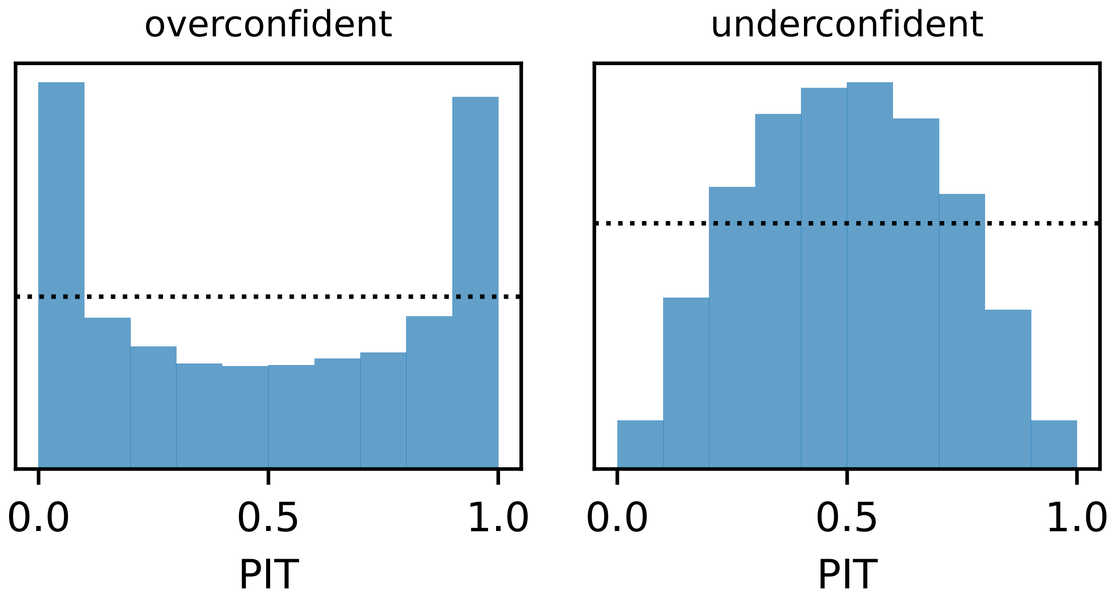
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necessary, **not sufficient** — *e.g.* prior would pass PIT and coverage

re-calibrating with PIT — if σ is uniformly too narrow or wide



coverage: the fraction of objects with PIT in [0.16, 0.84] should be 68%, etc

fit some constant α so that $|y - \mu| < \alpha\sigma$ corresponds to 68% of validation

much more on PIT and coverage later

scoring your regression

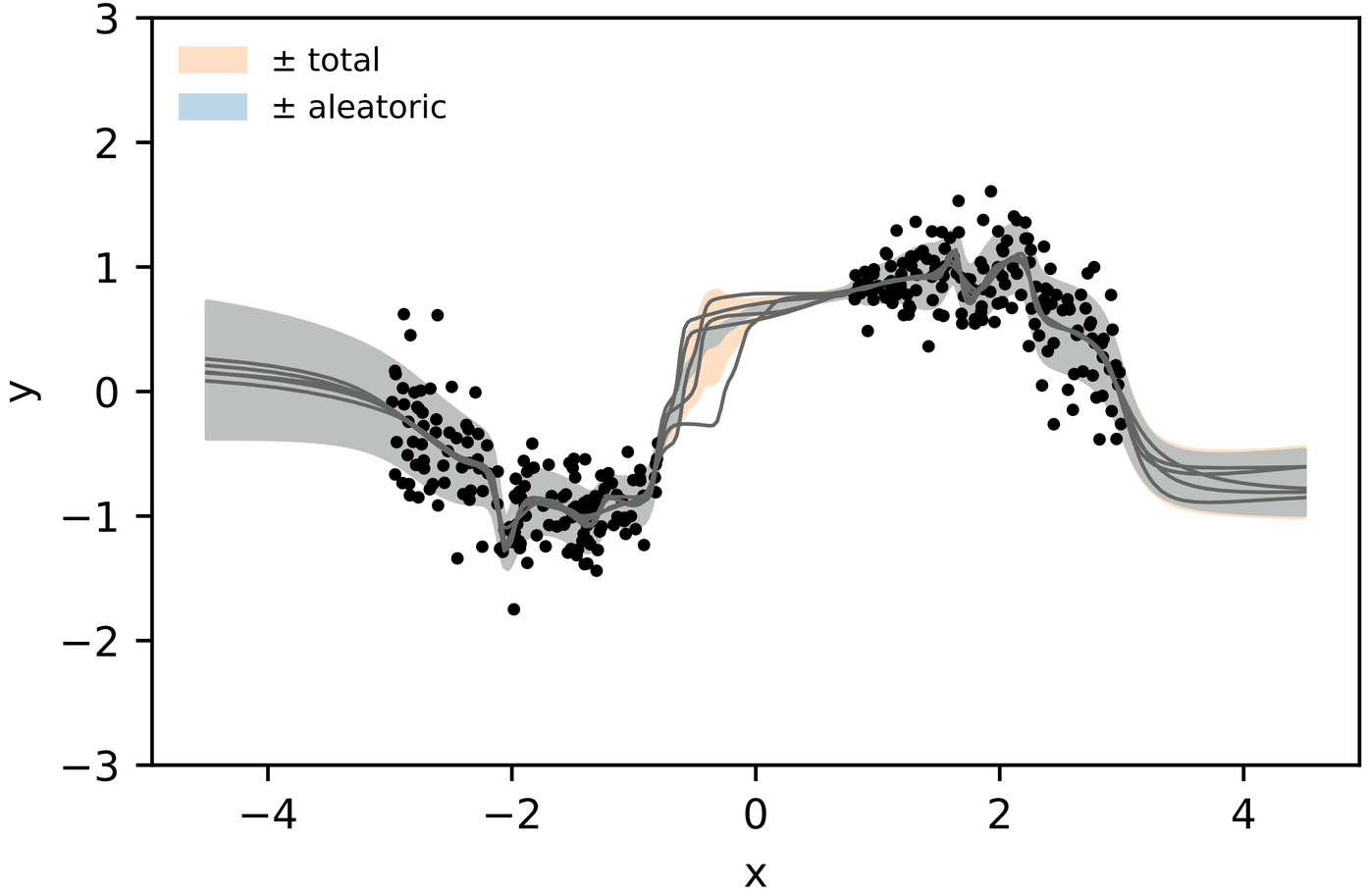
bias, RMSE	mean error and overall scatter
σ_{NMAD}	$= 1.4826\ \mathrm{median}(\lvert x_i - \mathrm{median}(x) \rvert)$
outlier fraction	catastrophic failures
test negative log-likelihood	a proper score
coverage, PIT	assessing the error bars

aleatoric versus **epistemic** uncertainty

aleatoric: intrinsic randomness — the scatter of y at fixed x. Property of the data; more training objects would not reduce it.

epistemic: lack of knowledge — what the training data didn't pin down. More training data would reduce it.

aleatoric versus **epistemic** uncertainty



5 networks trained on the same data

aleatoric versus **epistemic** uncertainty

aleatoric: intrinsic randomness — the scatter of y at fixed x. Property of the data; more training objects would not reduce it.

nothing you can do

epistemic: lack of knowledge — what the training data didn't pin down. More training data would reduce it.

dealing with **epistemic** uncertainty

ensembling (*Lakshminarayanan et al. 2017*): train M networks from different random starts; treat the prediction as their equal mixture.

MC dropout (*Gal & Ghahramani 2016*): keep drop out on at test time and average over many passes

in-class: photo-zs with uncertainties

ast384c/inclass/lec15_regress/photoz_uncertainty.ipynb